

Probability and Statistics (MT2005)

Midterm Exam

Date: July 8th 2024

Total Time: 2 Hours

Course Instructor(s)

Total Marks: 30

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Total Questions: 03

Roll No

Section

Student Signature

Attempt all the questions.

CLO 1: Describe the fundamental concepts in probability and statistics

Q1:

- (a) The following table summarizes the number of deceased beetles under autolysis (the destruction of a cell after its death by the action of its own enzymes) and putrefaction (decomposition of organic matter, especially protein, by microorganisms, resulting in

		Autolysis		
		High	Low	Medium
Putrefaction	High	14	8	21
	Low	17	12	0
	Medium	31	21	11

production of foul-smelling matter):

- What is the probability of the sample autolysis being low? [1 Marks]
- What is the probability of the sample autolysis being medium or putrefaction is high? [1 Marks]
- If the autolysis of a sample is high, what is the probability that the putrefaction is medium? [1 Marks]
- If the putrefaction of a sample is high, what is the probability that the autolysis is high? [1 Marks]
- Are low putrefaction and Medium autolysis mutually exclusive events? [1 Marks]

(b)

- How many three-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5, and 6 if each digit can be used only once? [1 Marks]
- How many of these are odd numbers? [2 Marks]
- How many are greater than 330? [2 Marks]

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CLO 2: Analyze the data and produce probabilistic models for different problems

Q2:

(a) Consider the following data,

129.2	185.3	218.1	182.5	142.8
155.2	170.0	151.3	187.5	145.6
167.3	161.0	178.7	165.0	172.5
191.1	150.7	187.0	173.7	178.2
161.7	170.1	165.8	214.6	136.7
278.8	175.6	188.7	132.1	158.5
146.4	209.1	175.4	182.0	173.6
149.9	158.6			

- (i) Find the five number summary ($\min, Q_1, Q_2, Q_3, \max$) of the above data [4 Marks]
 (ii) Construct Box-Whisker plot for the above data [1 Marks]

(b) Top Broadcast Shows. The viewing audiences, in millions, for the top 20 television shows, as determined by the Nielsen Ratings for the week ending October 26, 2008, are shown in the following table. Use cut point grouping with a first class of 12–under 13. Find the mean, median, mode and standard deviation as well. [5 Marks]

19.492	18.497	17.226	16.350	15.953
15.479	15.282	15.012	14.634	14.630
14.451	14.390	13.505	13.309	13.277
13.085	13.059	12.816	12.777	12.257

CLO 2: Analyze the data and produce probabilistic models for different problems

Q3:

(a) Suppose that the four inspectors at a film factory are supposed to stamp the expiration date on each package of film at the end of the assembly line. John, who stamps 20% of the packages, fails to stamp the expiration date once in every 200 packages; Tom, who stamps 60% of the packages, fails to stamp the expiration date once in every 100 packages; Jeff, who stamps 15% of the packages, fails to stamp the expiration date once in every 90 packages; and Pat, who stamps 5% of the packages, fails to stamp the expiration date once in every 200 packages. If a customer complains that her package of film does not show the expiration date, what is the probability that it was inspected by Jeff? [5 Marks]

(b) If a car agency sells 50% of its inventory of a certain foreign car equipped with side airbags, for the next 3 cars sold by the agency,

- (i) Construct the sample space with the help of tree diagram [2 Marks]
 (ii) Find the probability distribution of the number of cars with side air bag in tabular Form and formula form [3 Marks]